Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AI for Network Security

Course Code: C356

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	30 September 2026	Initial release of the AI for Network Security lesson plan — 2 days, 7.5 instructional hours per day (9:30 am – 5:30 pm), 4 topics aligned to LO1-LO4 and 12 hands-on AI-assisted labs.	Dr. Alfred Ang

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Course Information

Course Title	AI for Network Security
Course Code	C356
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7.5 instructional hours per day (15 hours)
Daily Timing	Day 1: 9:30 am – 5:30 pm · Day 2: 9:30 am – 5:30 pm (30-minute lunch, excluded from the 7.5 instructional hours)
Mode	Instructor-led, hands-on AI-assisted network security labs against authorised lab targets
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Identify and assess network threats and vulnerabilities by analysing ICT configurations.
- LO2: Evaluate organisational assets and develop risk management plans to address network infrastructure weaknesses effectively.
- LO3: Design and implement security measures for network components by managing security phases.
- LO4: Document security incidents and apply incident response configurations to manage risks based on established procedures.

Course Schedule

Day 1 — AI-Assisted Threat & Vulnerability Analysis · AI-Powered Risk Assessment and Management Planning

Time	Duration	Topic / Activity	Slides
9:30–10:00	30 min	Welcome, course introduction, ground rules and course outline	1-12
10:00–11:10	70 min	**Core concepts — network security and the CIA triad; why signature-based detection fails against zero-day, polymorphic and Living-off-the-Land attacks; where AI is applied in network security (concepts + demo)**	13-24
11:10–13:05	115 min	Topic 1 — AI-Assisted	25-44

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		Network Threat and Vulnerability Analysis: ICT networks and configuration, the threat landscape, pattern-of-life baselines, NDR and AI vulnerability prioritisation (concepts). Hands-on: Lab 1: Map the ICT Network and Analyse Its Configuration; Lab 2: Baseline Normal Traffic and Detect an Anomaly with AI	
13:05–13:35	30 min	Lunch break	
13:35–15:15	100 min	Hands-on: Lab 3: AI-Assisted Vulnerability Analysis and Prioritisation. Topic 2 — AI-Powered Network Risk Assessment and Management Planning: asset discovery and classification, CIA business impact (concepts)	45-59
15:15–17:30	135 min	Topic 2 continued — risk scoring on the 5x5 matrix, the four treatments, NIST CSF 2.0 and the NIST AI RMF (concepts). Hands-on: Lab 4: Build an Asset Register and Classify What Needs Protecting; Lab 5: AI-Assisted Risk Scoring and the Risk Register; Lab 6: Develop the Risk Management Plan. Day 1 recap and Q&A	60-76

Total instructional time: 450 minutes (7.5 hours), 9:30 am – 5:30 pm, excluding the 30-minute lunch.

Day 2 — Designing & Implementing AI-Enhanced Security Measures · AI-Assisted Incident Documentation and Response

Time	Duration	Topic / Activity	Slides
9:30–9:50	20 min	Day 1 recap and Q&A	
9:50–11:45	115 min	**Topic 3 — Designing and Implementing AI-Enhanced Network Security Measures: defence in depth and**	77-89

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		the three control categories; the planning, building and management phases; segmentation, zero trust and access control (concepts + demo)	
11:45–13:00	75 min	Topic 3 continued — AI-enhanced firewalls, IDS/IPS, UEBA and adaptive access control. Hands-on: Lab 7: Design the Layered Architecture Across the Three Security Phases; Lab 8: Implement AI-Enhanced Controls — Firewall Policy, IDS and Adaptive Access	90-101
13:00–13:30	30 min	Lunch break	
13:30–14:55	85 min	Topic 3 continued — securing the AI itself (NIST AI 100-2 attack taxonomy, NSA/CISA deployment guidance, MITRE ATLAS). Hands-on: Lab 9: Secure the AI Itself — Adversarial ML and the NSA/CISA Deployment Controls	102-108
14:55–16:50	115 min	Topic 4 — AI-Assisted Security Incident Documentation and Response: logging and audit design, SIEM/SOAR, AI alert triage, the NIST SP 800-61 lifecycle and incident documentation standards (concepts). Hands-on: Lab 10: Centralise Logs and Triage Alerts with AI; Lab 11: Document a Security Incident to the Standard	109-131
16:50–17:30	40 min	Hands-on: Lab 12: Implement the Incident Response Configuration and Playbook. Course recap, Q&A and course feedback	132-145

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Total instructional time: 450 minutes (7.5 hours), 9:30 am – 5:30 pm, excluding the 30-minute lunch.

Lab Reference (aligned to learning outcomes)

Topic / Learning unit	Weighting	Learning outcome	Labs	Slides
Topic 01: AI-Assisted Network Threat and Vulnerability Analysis	25%	LO1	Lab 1, Lab 2, Lab 3	Slides 34-49
Topic 02: AI-Powered Network Risk Assessment and Management Planning	25%	LO2	Lab 4, Lab 5, Lab 6	Slides 60-75
Topic 03: Designing and Implementing AI-Enhanced Network Security Measures	30%	LO3	Lab 7, Lab 8, Lab 9	Slides 90-107
Topic 04: AI-Assisted Security Incident Documentation and Response	20%	LO4	Lab 10, Lab 11, Lab 12	Slides 120-137

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